Domain and range and set notation

1

a $\{1, 2, 3, 4\}$ **b** $\{-7, -6, -5, -4\}$

c $\{1, 2, 3, 6, 7, 14, 21, 42\}$

2

a $\in$ **b** $\notin$ **c** $\in$ **d** $\notin$

e $\notin$

3

a False **b** True **c** True **d** True

e False **f** True **g** True

4 Example answers:
$\{x | x \text{ is an even number and } 2 \leq x \leq 8\}$
$\{x | x \text{ is a positive even number and } x < 10\}$

5

a

i $\{8, 1, 3, 9, 2\}$ **ii** $\{5, 3, 4, 1, 9\}$ **iii** Yes

b

i $\{2, 3, 6, 8\}$ **ii** $\{3, 8, 1, 7, 2\}$ **iii** No

c

i $\{4, 8, 2, 7, 3\}$ **ii** $\{4, 8, 9, 1\}$ **iii** Yes

d

i $\{1, 6, 3, 8, 2\}$ **ii** $\{3, 2, 7, 1\}$ **iii** Yes

e

i $\{7, 8, 5, 3\}$ **ii** $\{1, 9, 3, 2, 6\}$ **iii** No

f

i $\{2, 6, 9, 1, 5\}$ **ii** $\{6, 5, 1, 8, 3\}$ **iii** Yes

6

a

i $\{2, 4, 6, 8\}$ **ii** $\{2, 4, 6, 8, 10\}$ **iii** No

b

i $3 \leq x \leq 7$ **ii** $5 \leq y \leq 8$ **iii** Yes

c

i $2 < x < 7$ **ii** $u = 5$ **iii** Yes

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d
i $x = 5$ ii $2 \leq y \leq 6$ iii No

e
i $-9 \leq x \leq 9$ ii $-9 \leq y \leq 9$ iii No

f
i $-7 \leq x \leq 7$ ii $0 \leq y \leq 7$ iii Yes

g
i All real x ii $y \geq 0$ iii Yes

h
i $x \geq 0$ ii All real y iii No

7 $-6 \leq x \leq 6$

8
a
i All real values ii $\{y|y \leq 5\}$

b
i $\{x|x < 5\}$ ii $\{y|y > -2\}$

c
i All real values ii $\{y|y \geq 0\}$

9
a No **b** No **c** Yes **d** No

10
a No **b** No **c** No **d** Yes

11
a No **b** Yes **c** No **d** No

12 $\left[-\frac{1}{2}, \infty\right)$

13
a $\{y|y \leq 125\}$ **b** $\{y|0 \leq y \leq 125\}$

14
a No **b** Yes **c** No **d** No

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15

a Yes **b** No **c** No **d** No

Additional questions

16

a $\{1, 2, 3, 4\}$ **b** $\{1, 2, 3, 4\}$

17

a $\{1, 3, 6, 9\}$ **b** $\{2, 5, 7, 9\}$

18

a $\{-1, 3, 16, 19\}$ **b** $\{2, 5, 7\}$

19

a $\{-2, 0, 2, 4\}$ **b** $\{12, 51, 70\}$

20

a Possible answers:

- Domain: $-4 \leq x \leq 7$, Range: $-5 \leq y \leq 7$
- Domain: $[-4, 7]$, Range: $[-5, 7]$
- Domain: $\{x | -4 \leq x \leq 7\}$, Range: $\{y | -5 \leq y \leq 7\}$

b Possible answers:

- Domain: $-2 < x \leq 4$, Range: $-6 \leq y < 5$
- Domain: $(-2, 4]$, Range: $[-6, 5)$
- Domain: $\{x | -2 < x \leq 4\}$, Range: $\{y | -6 \leq y < 5\}$

c Possible answers:

- Domain: $-4 < x < 6$, Range: $-5 < y < 3$
- Domain: $(-4, 6)$, Range: $(-5, 3)$
- Domain: $\{x | -4 < x < 6\}$, Range: $\{y | -5 < y < 3\}$

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d Possible answers:

- Domain: $-1 \leq x \leq 3$, Range: $-1 \leq y \leq 6$
- Domain: $[-1, 3]$, Range: $[-1, 6]$
- Domain: $\{x | -1 \leq x \leq 3\}$, Range: $\{y | -1 \leq y \leq 6\}$

e Possible answers:

- Domain: $(-\infty, \infty)$, Range: $(-\infty, \infty)$
- Domain: All real values, Range: All real values

f Domain: All integer values, Range: All integer values

g Possible answers:

- Domain: $x \geq -5$, Range: All real values
- Domain: $[-5, \infty]$, Range: $[-\infty, \infty]$
- Domain: $\{x | x \geq -5\}$, Range: All real values

h Possible answers:

- Domain: $x = 4$, Range: $-6 < y < 6$
- Domain: $x = 4$, Range: $(-6, 6)$
- Domain: $\{x | x = 4\}$, Range: $\{y | -6 < y < 6\}$

21

a The domain is only $x \leq 8$

b The domain consists of all the real numbers

22

a The range consists of all the real numbers

b The range is only $y = 2$

23

a Domain: $35 \leq x \leq 50$
Range: $200 \leq y \leq 450$

b Domain: $0 \leq x \leq 17$
Range: $112 \leq y \leq 237$

c Domain: $0 \leq x \leq 134$
Range: $0 \leq y \leq 26.2$

24

a $x \geq 0$

b $y \geq 0.14$

25

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a 40 beats per minute

b 105°F

c The function output $f(10)$ is the average resting heart rate of a person at 10°F according to the model. However, since $T = 10$ falls well outside of the domain of the function, we do not have enough information to consider this a reliable estimate.

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